

Question 3:

Describe how to redesign the object to enable your main use case to be more efficient. Identify a performance target. What does it mean to be more efficient? For example: more efficient means my main use case will be _____ compared to the original design. Brainstorm! Show pros and cons for at least 3 different ways to redesign the object to meet this new target. Clearly show your work!

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Sienna's Response Q3

To improve the energy efficiency of my laptop during web browsing, I defined a clear performance target: complete the same research task while consuming less electrical power from the battery. In other words, a more efficient laptop performs the same computations, displays the same information, and provides the same user experience while requiring fewer watts to accomplish the task. Since the primary use case is internet browsing and online research rather than computationally intensive tasks such as gaming or video rendering, the redesign should improve energy efficiency without compromising portability, user comfort, or everyday usability.

My first concept revisited the research that originally inspired me to pursue engineering: piezoelectric computer keys. Every time a user types, mechanical energy is applied to the keyboard and immediately dissipated into the surrounding structure. My original idea was to embed piezoelectric ceramics beneath into a newly designed computer key to harvest this otherwise unused mechanical energy and convert it into electricity that could supplement the battery. The primary advantage of this concept is that the energy source already exists and is completely renewable. No additional user effort is required, and every keystroke contributes a small amount of electrical energy that would otherwise be lost. Unlike solar panels or external charging systems, the energy is generated directly from normal interaction with the device.

However, after two years of research and prototyping, I concluded that the power output of commercially available piezoelectric materials is far too small to significantly offset a laptop's energy consumption. Typical web browsing requires approximately 8–10 W of continuous power, whereas individual keystrokes produce only milliwatts of power for fractions of a second. Producing meaningful energy would require increasing the force required to press each key, increasing the size of the piezoelectric elements, or significantly altering the keyboard geometry. All three solutions negatively affect typing comfort, increase manufacturing complexity, and add weight to the device. Although this concept successfully demonstrated proof of concept, it ultimately became clear that harvesting user-generated mechanical energy could not provide meaningful improvements in overall system efficiency.

This naturally led to my second concept: rather than harvesting mechanical energy, could I recover the large amount of thermal energy continuously leaving the laptop? Every watt consumed by the processor ultimately becomes heat, so I investigated thermoelectric generators (TEGs) that convert temperature differences into electricity through the Seebeck effect. In theory, waste heat could be partially recovered and redirected back into the battery or power management system. The primary advantage of this concept is that it attempts to utilize energy already being generated during normal operation instead of allowing it to dissipate unused into the surrounding environment.

After reviewing recent literature, however, several limitations became apparent. Thermoelectric generators exhibit relatively poor conversion efficiencies at the modest temperature differences typically found inside consumer electronics. More importantly, the thermoelectric module itself introduces additional thermal resistance between the processor and the cooling system. This makes it more difficult for heat to escape, increasing processor temperatures and potentially causing earlier thermal throttling. In other words, recovering a very small amount of electrical energy may actually reduce the overall efficiency of the laptop by trapping significantly more heat inside the system. Although thermoelectric generators continue to show promise in high-temperature industrial applications and automotive waste heat recovery, current material limitations make them poorly suited for thin consumer laptops.

This realization fundamentally changed my approach to the problem. Instead of trying to recover wasted energy, I began asking how the laptop could be redesigned so that less energy is wasted in the first place. Since nearly all electrical energy consumed by the processor eventually becomes heat, reducing processor temperature can improve sustained performance while decreasing cooling requirements. This observation led to my final concept: redesigning the laptop chassis into a multifunctional thermal management component.

Current laptops already use heat pipes and vapor chambers to transport heat away from the processor toward a relatively small finned heat sink where a fan removes the heat through forced convection. The aluminum chassis participates only minimally in this process. Online I have also found that laptop enthusiast frequently demonstrate through "thermal pad" modifications that connecting the processor's cooling system directly to the aluminum chassis can reduce processor temperatures and improve sustained performance during prolonged workloads by allowing the entire enclosure to participate in heat dissipation. Numerous online thermal modification tests report sustained performance improvements of approximately 10–25% during processor-intensive workloads after increasing chassis participation in heat spreading.

Building on this concept, I propose redesigning the laptop chassis as a multifunctional thermal architecture rather than simply a structural enclosure. Instead of treating the aluminum chassis as passive structure, it becomes an engineered heat spreader that distributes processor heat across a much larger surface area. Increasing the effective heat rejection area lowers heat flux and reduces localized hot spots, allowing processor temperatures to remain lower during extended workloads. Rather than uniformly heating every external surface, high-conductivity pathways within the chassis would preferentially direct heat toward regions that users rarely touch, such as the rear hinge and side edges. High-contact regions including the palm rests and bottom center would remain thermally isolated through selective insulation or low-conductivity interface materials.

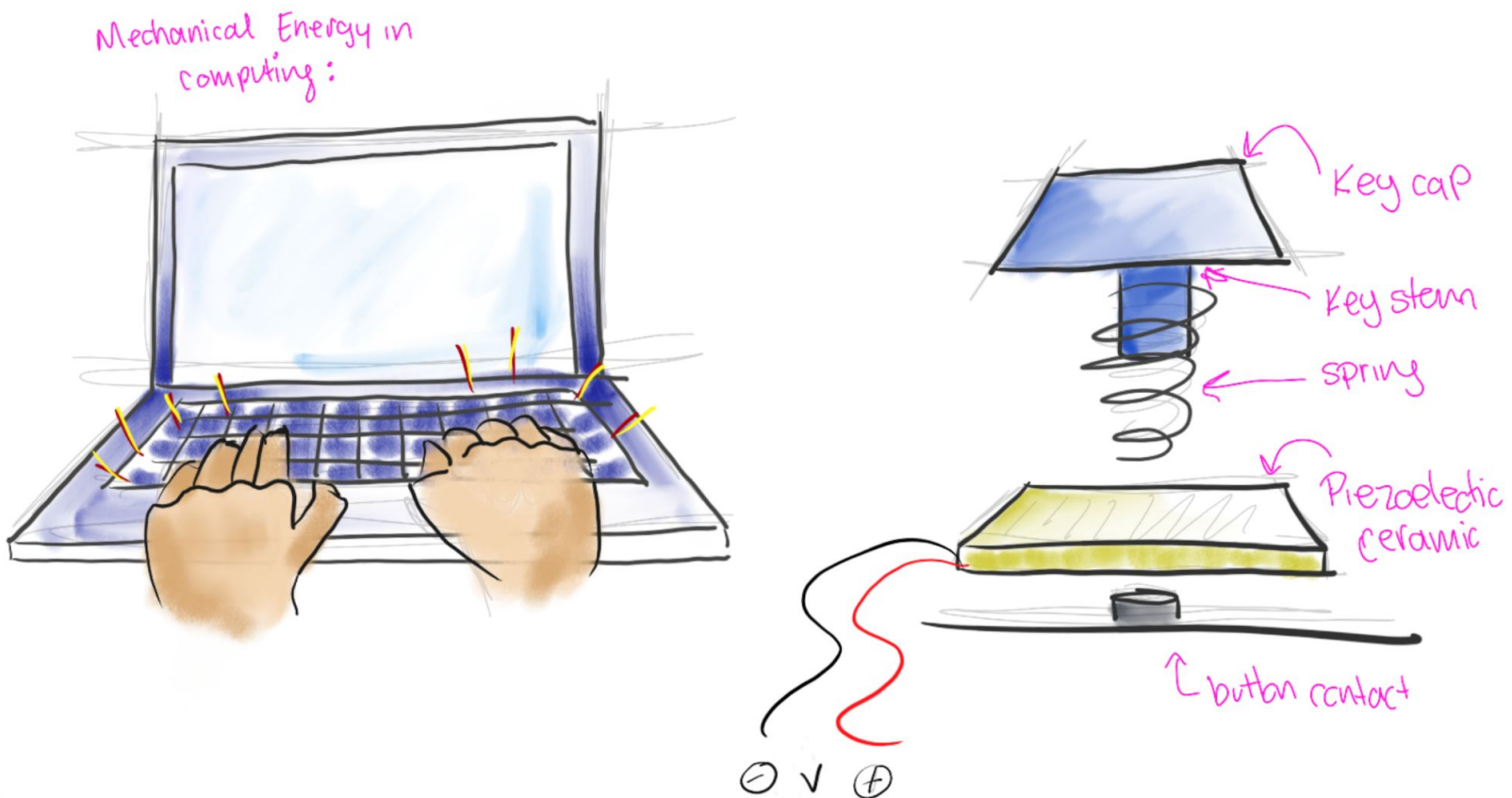
This concept offers several advantages beyond simply lowering processor temperatures. Lower operating temperatures reduce thermal throttling, allowing the processor to maintain higher clock speeds for longer periods while completing computational tasks more quickly. Faster task completion allows the processor to return to low-power idle states sooner, reducing the total energy required to complete a given workload. Additionally, improved passive heat spreading reduces reliance on active cooling fans, lowering both electrical power consumption and acoustic noise. Finally, integrating thermal management into an existing structural component aligns with modern product design trends emphasizing multifunctional engineering, where a single component performs several functions simultaneously instead of adding additional parts.

The primary challenges involve balancing thermal performance with user comfort while maintaining a lightweight, manufacturable product. Although spreading heat across the chassis lowers localized temperatures, users are sensitive to warm surfaces, particularly on the palm rests and bottom panel. Additional thermal pathways must therefore be carefully engineered to maximize heat rejection while maintaining acceptable touch temperatures. Manufacturing complexity, material selection, battery isolation, and long-term durability must also be considered before such a design could become commercially viable.

Overall, this progression—from generating energy, to recovering waste energy, to reducing energy waste altogether—reflects how my engineering thinking has evolved. Rather than asking how to produce more electricity, I ultimately concluded that the greatest opportunity lies in redesigning the laptop itself so it simply requires less energy to perform the same task. I believe this multifunctional thermal chassis provides the strongest balance between engineering feasibility, user experience, and meaningful improvements in laptop energy efficiency.

Idea 1. Piezoelectric Keyboard Energy Harvesting

Embed piezoelectric elements beneath each key to convert typing forces into electrical energy.



Pros:

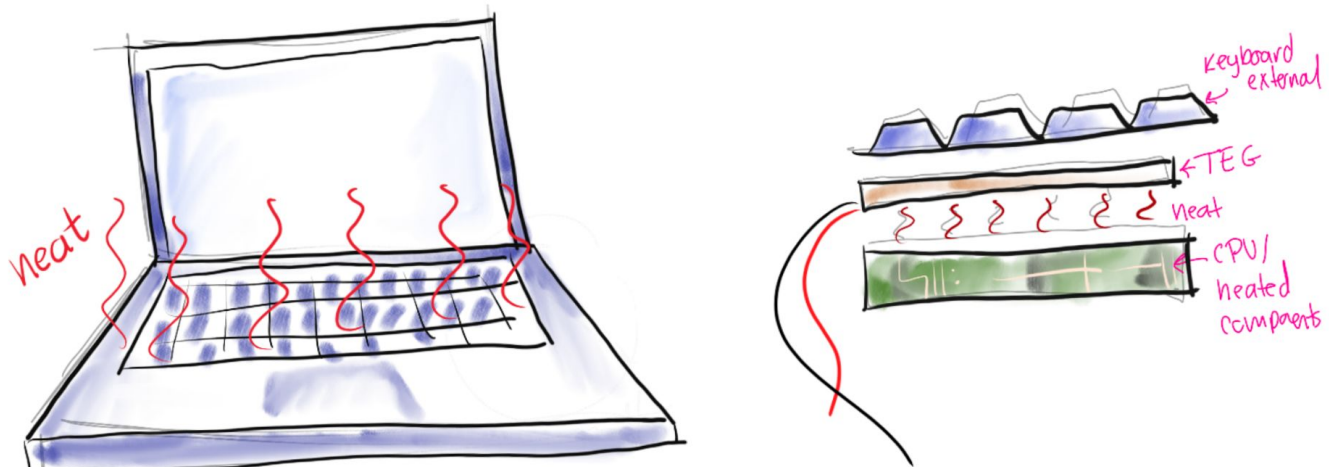
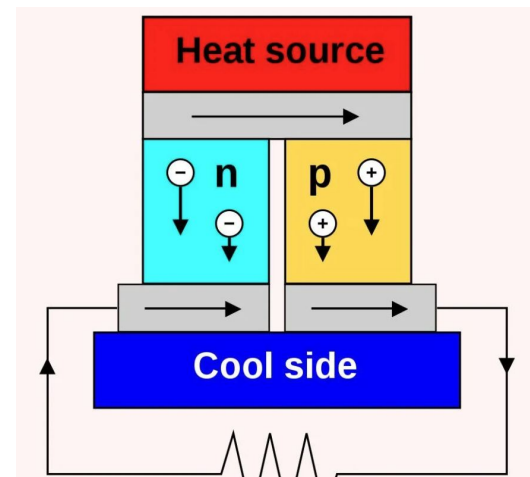
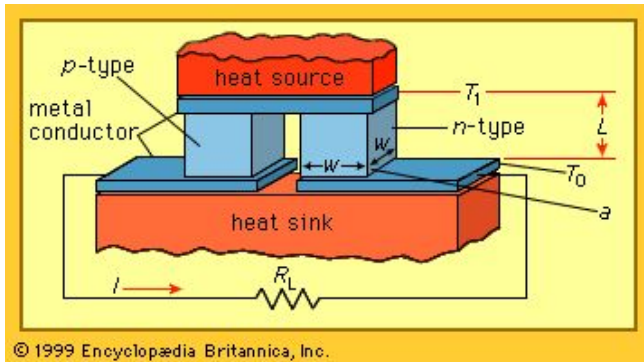
- Harvests renewable energy from normal typing
- Requires no additional user effort
- Builds on existing keyboard interaction
- Demonstrates proof of concept through my previous research

Cons:

- Extremely low power output (mW range)
- Negligible contribution compared to 8–10 W laptop demand
- May increase key stiffness and reduce typing comfort
- Adds manufacturing complexity, cost, and weight

Idea 2: Thermoelectric Waste Heat Recovery

Use thermoelectric generators (TEGs) to convert processor waste heat into electricity.



Pros:

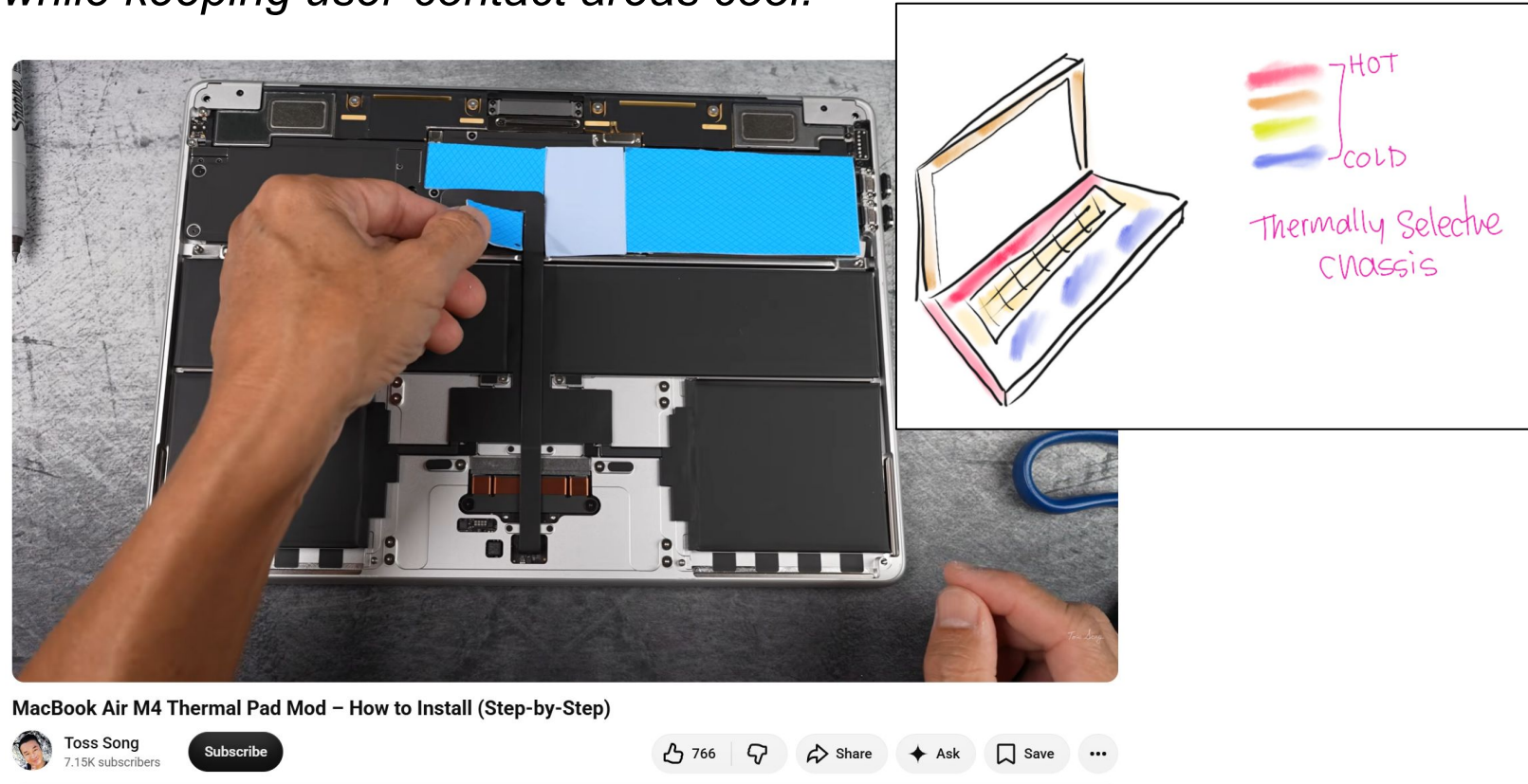
- Recovers energy that would otherwise be wasted as heat
- Operates continuously while the laptop is running
- Passive energy harvesting with no user interaction
- Existing technology used in some industrial applications

Cons

- Very low conversion efficiency at laptop temperatures
- Adds thermal resistance between CPU and heatsink
- Can increase processor temperatures
- May reduce overall system efficiency through thermal throttling
- Adds cost and integration complexity

Idea 3: Selective Thermal Architecture

Transform the aluminum chassis into an engineered thermal management system that routes heat toward low-contact regions while keeping user-contact areas cool.



Pros

- Reduces processor operating temperature
- Lowers fan usage and electrical power consumption
- Enables faster task completion and earlier return to low-power idle
- Improves sustained performance
- Uses existing materials and manufacturing methods
- Preserves thin laptop form factor

Cons

- Requires careful thermal pathway design
- Must balance cooling with user comfort
- Needs experimental validation before commercialization